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Tertiary geodynamical evolution of northwestern Greece: paleomagnetic results

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Paleomagnetic results have been obtained from 29 sites sampled in Lower to Upper Oligocene flysch sections of the Ionian Zone in northwestern Greece. They indicate a clockwise rotation of about 45° of the entire region. A comparison with previously published results relative to both younger and older formations shows that this rotation has occurred in two phases of comparable amplitude, one during Middle Miocene and the other during Pliocene and Quaternary, separated by a period of at least 7 Ma during which no significant rotation has occurred. No major rotation has occurred during the entire Paleocene and Eocene stages.

This large measured rotation implies a horizontal displacement of at least 100 km in Epirus. Possible mechanisms involving a rotation of the nappe sheets, rotation of the basement or a combination of the two are discussed.

The two phases of paleomagnetic rotation can be correlated with the Tertiary and Quaternary calc-alkaline volcanic activity in the Aegean Sea.

A palinspastic reconstruction shows that the orientation of the major structures, now striking N155°, was almost east-west in the Middle Miocene at the epoch of their formation.

1. Introduction

During the last three decades many important geological studies have documented the Tertiary evolution of the Hellenides and the collision of the Adriatic plate with the Rhodope massif. Although, the exact timing and the detailed mechanisms of the collision and of the post-collision evolution are still a subject of discussion, there is a general agreement that a large scale, westward directed thrusting occurred on Triassic evaporitic horizons, which resulted in the emplacement of nappe sheets over the Apulian platform. The last major phase of the "Alpine" activity in northwestern Greece occurred during Mid-Miocene [1].

Moreover, in the last 10 or 15 years, neotectonic and geophysical studies have provided new data and ideas about the most recent geodynamical evolution of the Aegean Area. Seismic studies, for instance, leave little doubt about the existence, in the southern part of the Aegean Arc, of a subduction zone with a rather well defined slab dipping 30–40° away from the trench and with associated back-arc volcanism.

Further, when moving from the southern to the northwestern portion of the arc there is a progressive transition to a zone of continental collision, evidenced by the gradual shallowing of the earthquake foci from 150–200 km under the Aegean Sea to the surface in the Ionian Islands region. Neotectonic studies have also shown that the compressional regime obtained today from focal mechanisms and in-situ constraint measurements [2] has existed in the region of the Ionian Islands throughout the Pleistocene and the Pliocene and started at the Mio-Pliocene boundary, when the present active margin most probably appeared [3–5]. Although the period between the end of the

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Alpine history and the beginning of the activity of the present arc has been less intensively studied, geological data show that a carbonate platform subjected to extensional tectonics existed in the region of today's arc [3,5]. The former compressive margin was situated more to the east in the Ionian Zone.

In contrast with this impressive body of geological data, Greece has been largely neglected from a paleomagnetic point of view, until very recently. Before 1979 [6], only an isolated result from the Almopias volcanics [7] and a questionable comparison of the relative orientation of the Pindus ophiolites and of the Argolis Peninsula [8] had been published.

New ideas, which led to the development of several models for the geodynamical evolution of the Aegean Area, certainly also contributed to the renewed interest evidenced by a number of recent paleomagnetic studies. Results from the Upper Miocene sections in Crete and Rhodos [9] have shown that no significant rotation of the southeastern part of the arc has occurred since the Late Tortonian. On the other hand, results from the Ionian Islands [5,10] yield evidence of 25° clockwise rotation with respect to the present geographic pole since the Mio-Pliocene boundary. While no significant rotation of this region was detected between the Serravallian-Tortonian and the Mio-Pliocene boundaries, preliminary results from 16 Oligocene flysch sections in the Ionian Zone [11] indicate an earlier phase of some 25–30° additional clockwise rotation. Also, Horner and Freeman [12] obtained precise paleomagnetic results from 51 sites sampled in the Paleocene and Eocene "microbreccious limestones" of the Ionian Zone which indicate a 38° clockwise rotation of this region since Upper Eocene.

In this paper, we report additional paleomagnetic results obtained from lower to Upper Oligocene flysch sections in the Ionian Zone, both in Epirus and Akarnania. A comparison of these results with the above-mentioned paleomagnetic results within the framework of known geological data then allows a fairly complete description of the geodynamical evolution of northwestern Greece for the entire Tertiary and Quaternary periods.

2. Choice of sampling sites

The Ionian Zone is one of the four "isopic zones" into which Brunn [13] divided the orogenic belt of the external Hellenides on the basis of their Mesozoic facies development. It extends from Epirus, where three subzones may be distinguished, to the Peloponnesos.

Its tectonic setting varies considerably from north to south. In Epirus large, faulted folds dominate, at least as far as the formations situated above the huge mass of evaporites are concerned. In Akarnania, the amplitude of the folds is smaller and low-angle unconformities are often encountered.

Taking these complex structures into account, sites were only sampled in the flanks of major structures with horizontal axes aligned along the main direction of the belt. Regions involving small complex structures with non-horizontal axes and no direct regional significance, such as the Igoumenitsa area in Epirus or the portion of Akarnania west of a line Amfilokia-Aitolikon were thus avoided. Indeed bedding correction techniques for plunging axes give a unique result only when the detailed tectonic history is known. On the other hand, 30 sites were sampled in the Botzara syncline because this is a major structure of northwestern Greece and any data about its rotation would be highly significant for the entire region. Moreover, the axis of this syncline is very nearly horizontal all along its length, so that even if its formation is due to more than one tectonic event, folding has occurred about the same axis, and the usual tilt correction can be safely applied to the paleomagnetic results. For the same reason, we have sampled 12 sites in another major structure, the Epirus-Akarnania Syncline.

On a local scale, sites were selected on the basis of sedimentary and magnetic criteria. In general fine-grained sediments were chosen preferentially because their sedimentation rate is usually low enough to average out completely the secular variation of the geomagnetic field over the thickness of a sample and because this kind of sediment is usually deposited in calm water. Sections close to major faults or containing slumped beds or landslides were avoided.

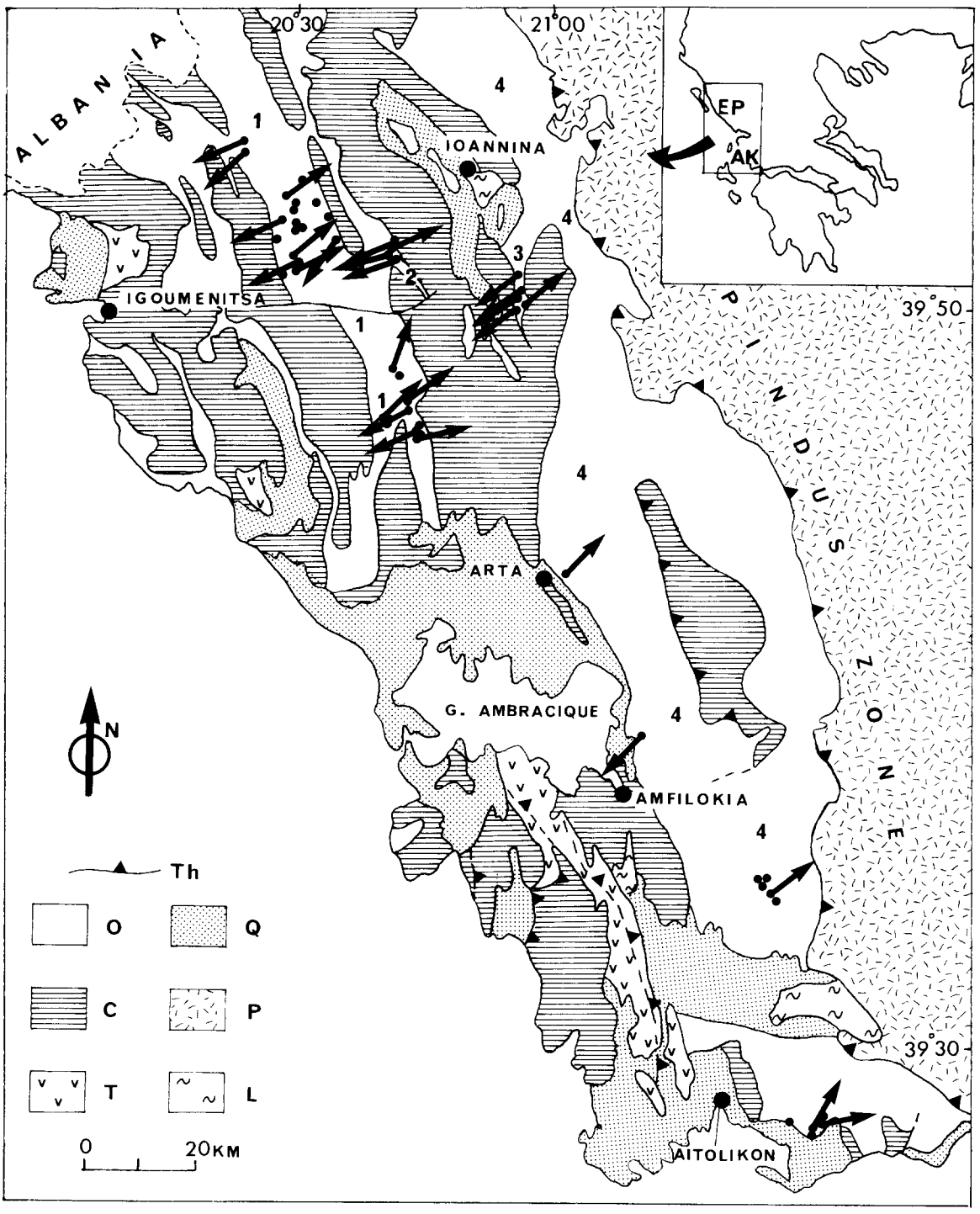


Fig. 1. Location of the sampling sites and results obtained in Ionian flysch sections (Epirus and Akarnania). *Q* = Quaternary deposits; *O* = Oligocene and Lower Miocene detritical formations; *C* = Jurassic, Cretaceous and Eocene limestones; *T* = Triassic evaporitic formations; *P* = Pindus massif; *L* = lakes. Main synclines: *1* = Botzara; *2* = Dodoni; *3* = Ellinikon; *4* = Epirus-Akarnania, *Th* = main thrusts.

Careful attention was paid to the colour of the sediments and only blue-grey marls were sampled. Laboratory studies have shown that sediments of this colour have not been affected by weathering, so that they are free of any chemical remanent magnetism. Even though this colour may not be original and may result from reduction phenomena after deposition (D. Sorel, oral communication), we have assumed that the "magnetic age" is not significantly different from the stratigraphic age.

The final choice of outcrops was made directly in the field using a portable spinner magnetometer [14]. Many of the geologically suitable formations have a very low level of magnetization, difficult to measure during stepwise demagnetization, even with the cryogenic magnetometer which we have used during the second half of this work. Because sites are not usually situated along the road and are difficult to sample, whenever two or more suitable and geologically equivalent sites were located in a given zone, only the one with the highest magnetization intensity was sampled. Sites with low magnetization level were however collected in regions where it was impossible to locate better magnetized formations.

Drilling and orientation of the cores was done with standard paleomagnetic techniques, using compressed air to cool the corer and remove the dust. More than 1200 cores were obtained from 51 sites in Oligo-Miocene sections in the Ionian Zone.

The location of the sampled sites is shown in Fig. 1.

3. Results

3.1. Biostratigraphic determinations

The precise stratigraphic position of the sampled sites has been obtained by biostratigraphic dating based on nannofossil associations.

In Epirus all the sites sampled in the Radhovizi Formation were assigned either to NP 25 or to NP 24/25 Zones and are thus of Upper Oligocene age. This formation was formerly considered as Aquitanian. All the other age determinations in Epirus agree with the results given by the IGRS-IFP [1].

On the other hand, all the sites sampled in Akarnania were found to belong to the NP 24/25 boundary zone, in contrast with previous determinations by the British Petroleum [15] and Piper et al. [16]. Indeed these authors have denied the presence of Oligocene flysch in Akarnania and ascribed the presence of Oligocene fauna in their sections to large-scale erosion and reworking during the Miocene. Their point of view has already been strongly questioned by Bizon [17] on the basis of well established correlations with Oligocene sections in Epirus and Peloponnesos. Our present determinations thus lend support to Bizon's [17] interpretation.

The stratigraphic position of the investigated sites is indicated in Table 2, along with the paleomagnetic results. It can be seen from this table that we have not succeeded in locating any paleomagnetically suitable outcrop of Lower to Middle Miocene age (between about 24 and 12 My).

3.2. General sedimentary and magnetic properties

Although careful attention was paid to sample fresh outcrops, X-ray analyses were systematically performed on samples from the different sites. The obtained spectra, five of which are reported in Fig. 2, show that the mineralogy is dominated by the three detrital clays chlorite, illite and kaolinite. Quartz and calcite are also present in variable amounts. These results thus confirm that no significant weathering has affected the sampled formations.

Because the sampled flysch formations are of turbiditic origin, the magnetic fabric has been carefully examined by measuring the anisotropy of all the samples with a laboratory modified Digico Anisotropy Delineator and using the calibrating procedure given by Veitch et al. [18]. A typical result from a site in Epirus is shown in Fig. 3. The magnetic fabric of a great majority of the samples is characterized by an oblate susceptibility ellipsoid with a rather high degree of anisotropy ($K_{max}/K_{min} = 1.10$) and a mean axis of minimum susceptibility, K_{min}, almost exactly perpendicular to the bedding plane ($\bar{I} = 85°$) while K_{max} and K_{int} very closely lie in the plane. This is an

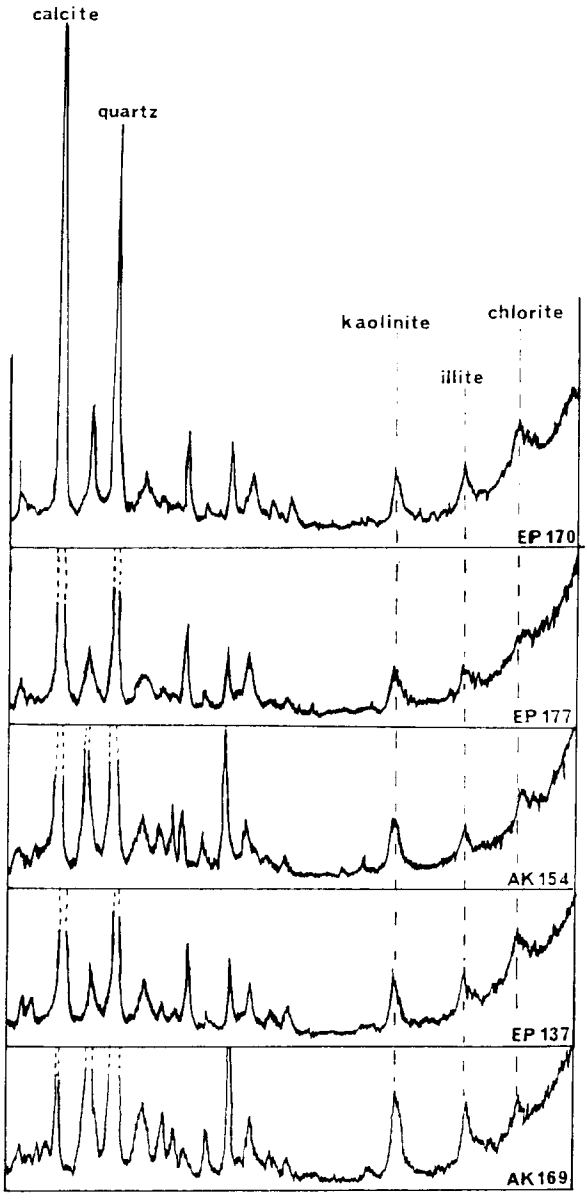


Fig. 2. Five spectra of X-ray analyses showing that the sampled sediments are composed essentially by three detrital clays.

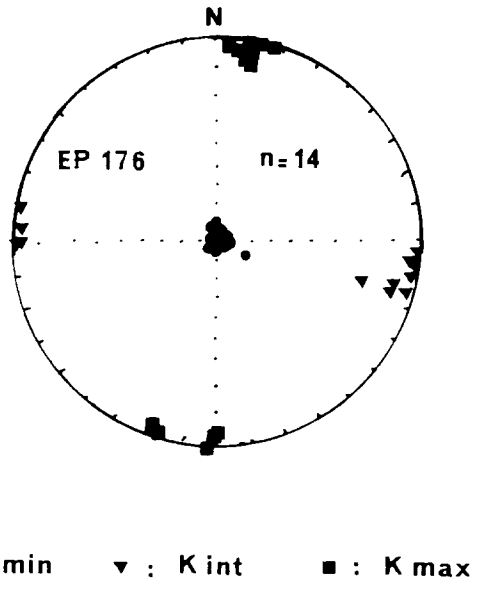


Fig. 3. Stereographic projection of the direction of principal susceptibility axes of site EP 176 which represents a typical result, except for the southern part of the Epirus-Akarnania Syncline (see text).

indication of a planar sedimentary fabric, furthermore characterized by a moderate magnetic lineation ($K_{max}/K_{int} = 1.04$), with K_{max} directed in a roughly north-south direction in most of the sites. This trend of the magnetic lineation will be discussed in a future publication. For the present purposes, these results provide a good evidence of a calm and subsequently undisturbed deposition of the sediments. For the three sites of a syncline in southern Akarnania, it was found that this planar magnetic fabric is disturbed, with the K_{min} axes not exactly perpendicular to the bedding plane and the susceptibility ellipsoid more prolate than oblate. Although these sediments appear from visual inspection to be completely undisturbed, these characteristics of the susceptibility ellipsoid could indicate the existence of moderate tectonic strains in this syncline [19,20]. The rather high scatter of the remanent magnetization directions observed in these three cases is probably related to this phenomenon. The mean paleomagnetic direction from these sites was, however, quite coherent with that from the other ones.

Different techniques were used to identify the magnetic carriers. First of all, the IRM acquisition in fields up to 1.5 T was studied for at least two samples per site. Typical results, shown in Fig. 4 indicate that magnetite is the main magnetic carrier. Only in five cases a continuous increase in the

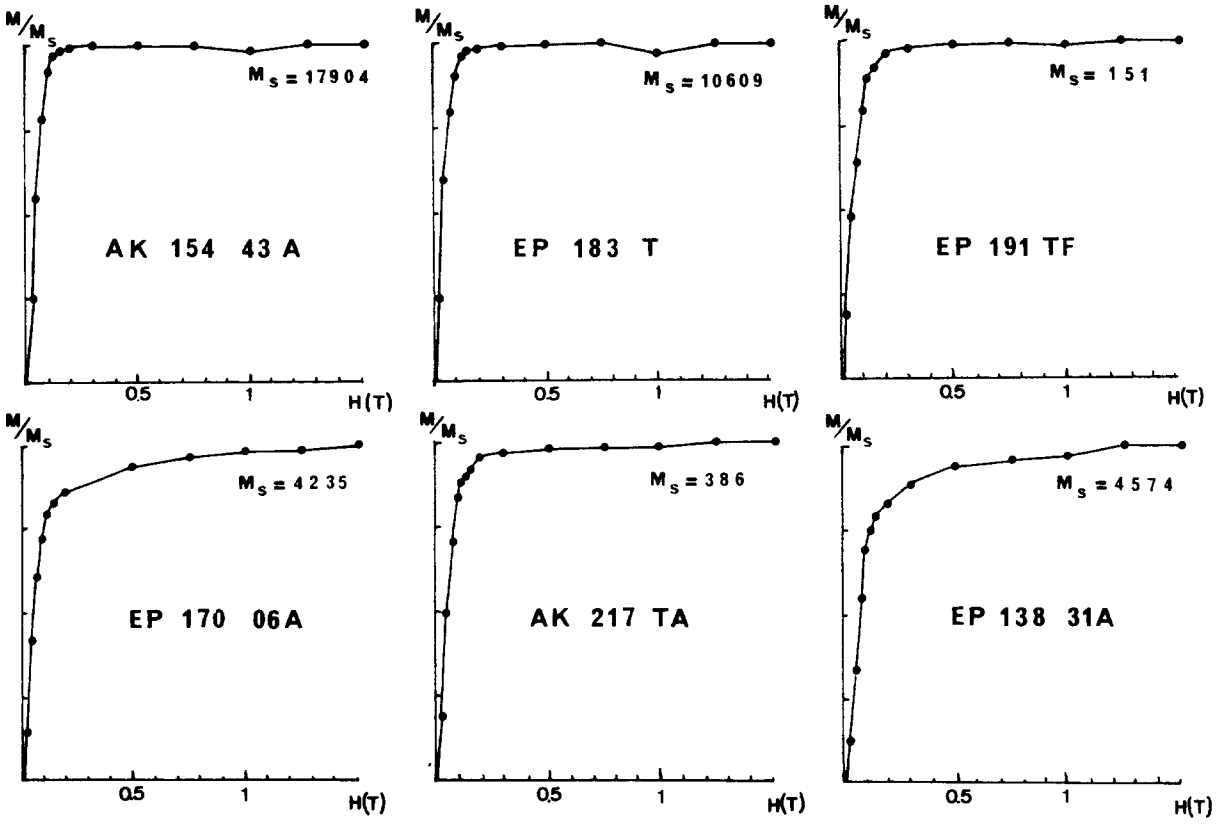
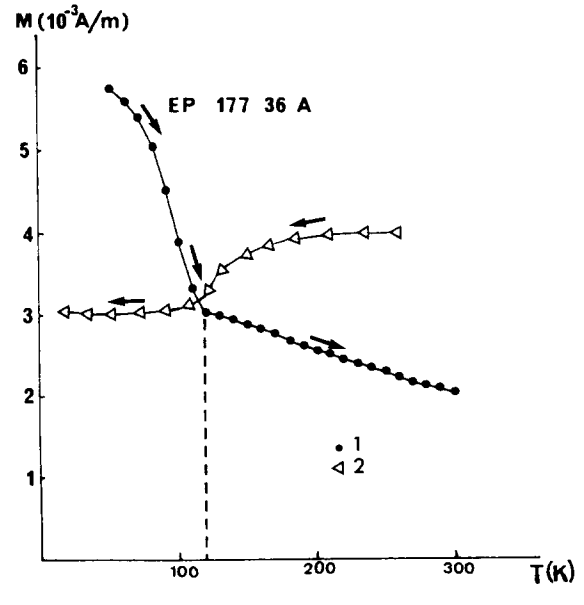


Fig. 4. IRM diagrams of 6 samples from different sites from the Ionian flysch sections. M_s is given in 10^{-3} A/m.



IRM for fields higher than 0.2 T indicates the presence of a magnetic mineral of higher coercivity. In all cases, however, the AF median destructive alternating field of the SIRM was less than 0.3 mT, confirming the dominant presence of magnetite. Moreover the presence of the Verwey transition at 120°K, corresponding to multidomain magnetite, has been clearly established in one sample from site EP 177 by P. Rochette (Fig. 5).

The remanent magnetization of the samples was measured using either a Digico spinner magnetom-

Fig. 5. Temperature dependence of the IRM of a sample from the site EP 177 (Botzara Syncline). 1 = heating of IRM acquired at 50°K in 3 T; 2 = cooling of IRM acquired at 300°K in 3 T. The change in the slope at 120°K corresponds to the Verwey transition. Assuming that no other impurities are present this temperature is an indication of a titanomagnetite with a low titanium content [41].

eter or a LETI three-axis cryogenic magnetometer. The NRM intensity of the sampled sites ranges from 75×10^{-3} to 3×10^{-3} A/m. Differences up to a factor of 10 within-site and over 100 between sites were observed. No clear correlations between the intensity of the NRM and the geographical or stratigraphical position of the samples have, however, been obtained with the very important exception of the Burdigalian sites. Indeed, these sites were all weakly magnetized and difficult to measure after the first steps of thermal or AC demagnetization, in spite of considerable care and of the double precision measuring procedure of Vandenberg [21] which involves four independent measurements of each samples. No reliable result has in fact been obtained from Burdigalian formations.

Stepwise thermal demagnetization of all the samples, involving a minimum of 12 steps from room temperature up to at least 550°C, has been used throughout this study, because the obtained results proved to be more consistent and less scattered than those obtained by AF treatment, although the mean values were the same in the two cases.

Changes in the magnetic susceptibility of at least two samples per site were measured at each step of the thermal demagnetization using a susceptibility bridge. As shown in Fig. 6 the susceptibility is thermally very stable as no changes of more than a factor of two have been observed between room temperature and 500°C. The magnetic minerals are not seriously affected by the thermal treatment.

Typical Zijderveld demagnetization diagrams are shown in Fig. 7. Apart from a small viscous component removed at 200–250°C, all the samples have a single stable component of magnetization whose direction can be determined precisely by hand fitting a straight line through, at least, the last 5 or 6 points. Within-site scatter is quite small as shown by the stereographic projections of Fig. 8. When samples of both polarities are present in the same site, their paleomagnetic directions are exactly opposed, which is an indication of good magnetic stability and satisfactory magnetic cleaning.

In both the Botzara and the Ellinikon synclines only one site meeting the sampling requirements was located on the eastern flank which is much steeper and thus less favorable for outcrops than the western one. Two fold tests comparing these two sites to the two closest sites on the western flank are positive even at the 1% significance level

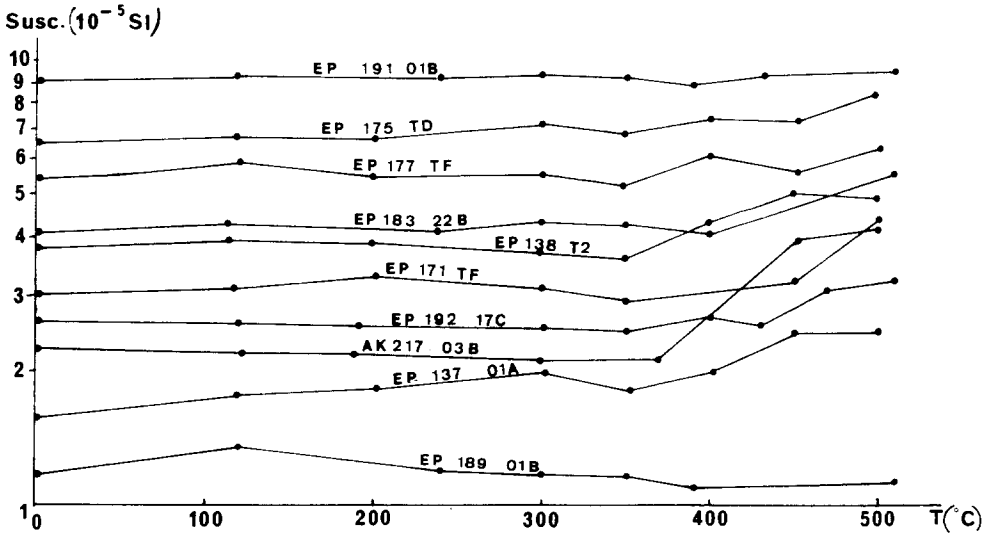


Fig. 6. Evolution of the susceptibility of samples during stepwise thermal demagnetization.

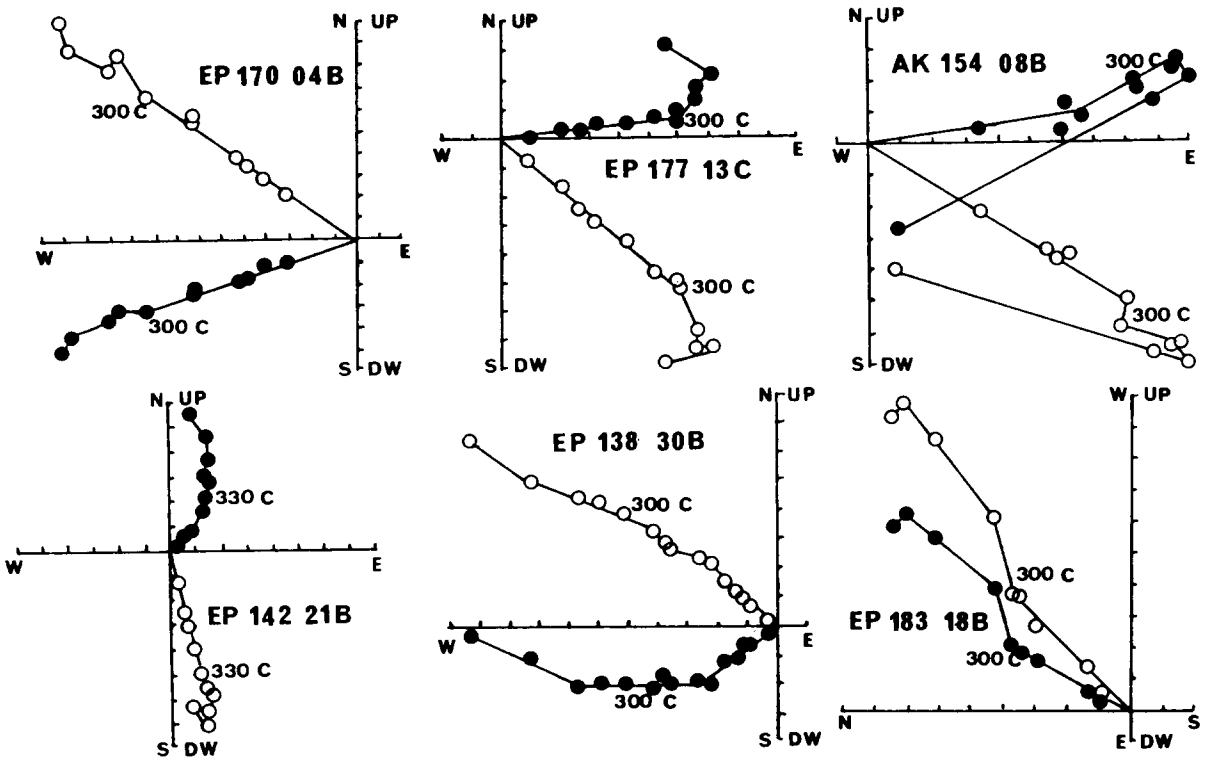


Fig. 7. Thermal demagnetization plots of various samples from sites of Ionian flysch. The number of steps depends on the samples (9–15) and the maximum temperature varies between 550 and 600°C. Open circles: projection on east-west or north-south vertical plane; full circles: horizontal projection.

[22]. Furthermore scattering of data obtained from the 29 sites yielding reliable results is significantly smaller after bedding correction (Table 1). Thus the fold tests support the interpretation of stable paleomagnetic directions predating the folding of the investigated structures.

In summary, the laboratory studies and field tests yield evidence that the investigated formations are well suitable for a reliable paleomagnetic study.

3.3 Paleomagnetic results

The results obtained from 29 sites sampled in Lower to Upper Oligocene flysch sections are reported in Table 2 and also represented on a stereographic projection in Fig. 9. When all the reverse sites are inverted through the origin, the paleomagnetic direction obtained by Fisher's statistics is:

$N = 29 \quad D = 58.0° \quad I = 38.8° \quad k = 30$

$\alpha_{95} = 4.8°$

TABLE 1

Fold test obtained from two synclines in Epirus and comparison between scattering of the data from the 29 sites before and after bedding correction

Botzara Syncline		Ellinikon Syncline
$N = 22$		$N = 24$
$(k_2/k_1) = 5.5$		$(k_2/k_1) = 2.4$
$F_{(42,42)}\ 1\% = 2.07$		$F_{(46,46)}\ 1\% = 2.0$
All sites:		
$N = 29$	$k_2/k_1 = 4.3$	$F_{(56,56)}\ 1\% = 1.9$

N: total number of samples of the two sites taken in opposite flank in each structure. k_1 and k_2: precision parameter of the Fisher's statistic. F: limiting value derived from F tables at the 0.99 significance level.

TABLE 2

Paleomagnetic results from 29 sites of Ionian flysch. The sites are grouped according to their structural position. The datation is based on nannofossils and the average is calculated by usual Fisher's statistic

Site	Age	n/N	D_{nb}(°)	I_{nb}(°)	D_b(°)	I_b(°)	k	α_{95}(°)
Ellinikon								
EP 137		11/11	47.5	25.4	47.5	25.4	48	6.1
EP 138		9/11	241.5	−36.7	239.0	−34.7	157	3.7
EP 139		3/4	61.0	30.3	56.8	31.4	–	–
		13/14 } 16/18	243.7 } 236.5	−33.6 } −35.5	231.5 } 239.0	−36.0 } −35.0	281 } 114	2.3 } 3.3
EP 140		2/2	42.3	43.4	50.0	32.0	–	–
		8/8 } 10/10	223.0 } 223.5	−45.3 } −45.6	231.5 } 231.0	−34.0 } −33.8	237 } 286	3.2 } 2.6
EP 141		15/15	234.0	−27.7	236.0	−35.0	244	2.3
Dodoni	Lower							
EP 170	Oligocene	8/10	250.0	−22.0	250.0	−32.0	107	4.5
EP 171	NP 21	10/10	251.3	−27.5	256.0	−27.7	100	4.4
EP 172		6/6	62.5	35.6	62.5	35.6	61	7.3
		4/4 } 10/10	262.0 } 70.0	−39.0 } 37.0	262.0 } 70.0	−39.0 } 37.0	83 } 43	7.6 } 6.7
EP 192		10/10	253.0	−23.5	249.0	−27.0	68	5.3
EP 216		8/11	39.0	46.2	44.5	29.6	16	12
Botzara								
EP 197	Middle Oligocene?	8/11	217.3	25.6	217.0	−27.8	33	8.6
EP 173	Middle	11/11	243.0	−27.5	243.0	−27.5	24	8.6
EP 174	Oligocene	8/8	214.2	−42.5	227.5	−40.7	12	14
EP 185	NP 24	8/10	350.6	58.4	43.0	45.0	32	8.7
EP 143		4/4	48.3	66.0	58.0	44.0	190	5.0
		9/11 } 13/15	233.6 } 235.5	−68.6 } −68.6	242.5 } 241.3	−46.0 } −45.4	155 } 167	3.7 } 3.0
EP 175	Middle	6/6	9.7	69.3	45.7	56.0	21	12
	to Upper	5/5 } 11/11	196.0 } 13.1	−68.5 } 69.0	228.0 } 47.0	−54.0 } 55.0	78 } 34	7.0 } 7.2
EP 176	Oligocene	9/9	57.2	75.5	59.0	45.6	53	6.4
	NP 24/25	5/5 } 14/14	226.1 } 53.0	−75.3 } 75.5	235.0 } 57.5	−45.6 } 45.6	47 } 54	9.0 } 5.0
EP 189		6/6	179.0	82.0	65.0	49.0	222	4.0
		4/8 } 10/15	148.3 } 213.2	−82.1 } 86.7	224.0 } 56.8	−44.0 } 48.0	– } 72	– } 5.5
EP 190		11/14	221.0	−46.0	240.0	−55.0	19	9.7
EP 142	Upper	11/15	37.6	71.4	51.2	37.4	97	4.3
EP 177	Oligocene	6/6	60.5	75.7	59.6	51.7	62	7.2
	NP 25	8/8 } 14/14	257.3 } 267.6	−76.5 } −76.3	250.0 } 246.0	−53.4 } −53.0	77 } 67	5.6 } 4.6
EP 183		9/11	245.0	−65.5	247.0	−45.7	25	8.5
EP 186	Upper	1/4	264.7	75.0	81.0	44.0	–	–
	Oligocene	8/9 } 9/14	100.0 } 275.8	−71.8 } 72.3	254.0 } 76.0	−46.0 } 46.0	20 } 22	12 } 11
EP 187		10/10	60.9	31.0	20.0	37.0	28	8.4
Akarnania								
AK 217	Upper Middle Oligocene NP 24/25	5/12	250.0	−33.3	247.0	−27.0	20	14
AK 169	?	5/6	1.7	75.3	52.0	37.0	62	7.9
AK 154	Upper	7/10	78.0	44.0	75.3	26.4	30	9.6
AK 155	Oligocene	12/14	26.7	34.5	30.7	27.0	27	7.7
AK 156	(?)	7/11	81.4	20.3	81.4	20.3	25	10

n/N: number of samples selected at the 2σ level/number of samples measured. Sites containing samples of both polarities are indicated and the polarity assigned to these sites is the one of the majority of samples. The paleomagnetic directions of samples which were in minority were inverted through the origin and averaged with those of the other samples. The directions are reported before (D_{nb}, I_{nb}) and after (D_b, I_b) bedding correction. The result of EP 177 is slightly different from the one previously reported in Kissel et al. [11] because of better appreciation of the bedding plane

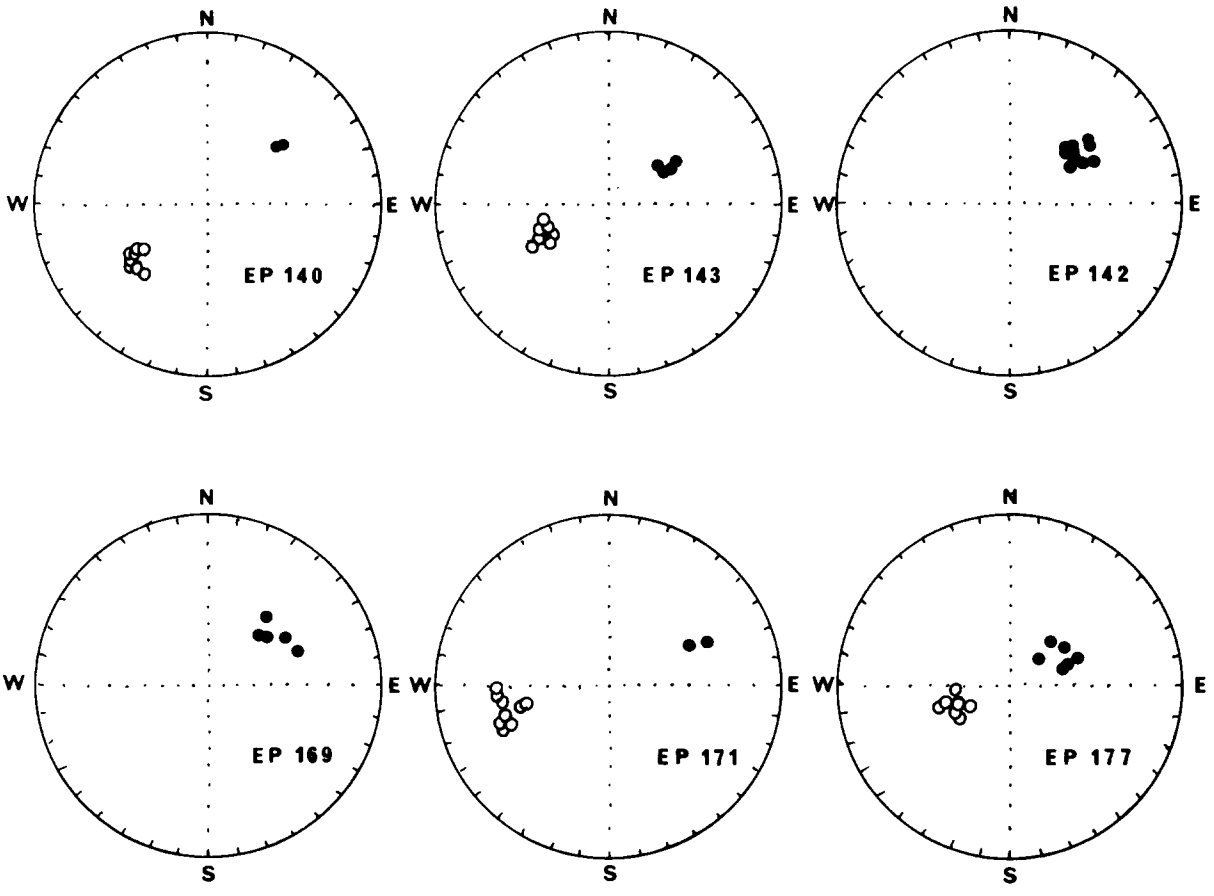
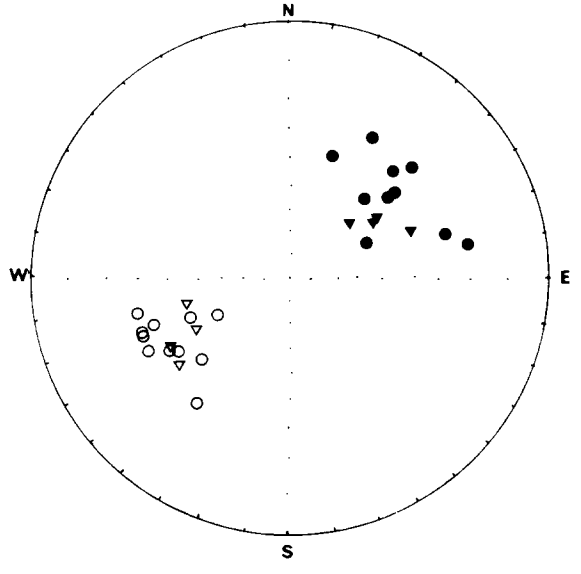


Fig. 8. Stereographic projections of results from 6 sites of Ionian flysch Full circles: normal samples, open circles: reverse samples.



in agreement with previously reported preliminary results [11].

4. Discussion

In spite of considerable amount of work which has been dedicated to the problem of the relative

Fig. 9. Stereographic projection of the paleomagnetic directions of Table 1 (29 sites). Full circles: sites containing only normal samples; open circles: sites containing only reverse samples; full and open triangles: sites containing both polarities with, respectively, a majority of normal samples and a majority of reverse samples. For the sites containing samples of both polarities, the paleomagnetic directions of samples which were in minority were inverted through the origin and averaged with those of the other samples.

movement of Europe and Africa, wide differences still exist among different authors about the geodynamical evolution of the periadriatic regions. Channel et al. [23], for instance, consider that the Apulian platform has behaved as a promontory of the African plate and thus has moved with Africa since the Early Mesozoic. Vandenberg, on the contrary, suggested that an anticlockwise rotation of 27° of Apulia relative to Africa occurred in the Early Tertiary [21]. According to the most recent global reconstitution of the history of the Mediterranean [24], the Apulian platform has undergone an anticlockwise rotation of 25° relative to Africa 110 My B.P. and has moved in coordination with Africa since then.

Thus, in spite of these discrepancies, no major movement seems to have occurred between the Apulian platform and Africa since Middle Tertiary.

It thus seems reasonable to refer our data to the African Oligocene paleomagnetic pole determined by Schult [25] also included in a recent polar wander path for Africa published by Tauxe et al. [26]. On this basis, the sampled sites, reduced to a mean geographic location of (39.5°N, 20.5°E), are expected to have a paleomagnetic direction defined by ($D = 8.5°$, $I = 45°$).

The mean paleomagnetic inclination obtained from the flysch sections is 6° shallower than expected. This could arise from errors in the paleomagnetic pole. However, an inclination error of the same order of magnitude has been observed in Upper Miocene and Pliocene marls from the external sedimentary arc and attributed to compaction of the sediments [5]. The same phenomenon could thus account for the discrepancy observed here.

The mean obtained declination implies a clockwise rotation of $49.5 \pm 6.5°$ since Lower Miocene. The absence of any trend in the paleomagnetic data indicates that no significant rotation has occurred during the entire Oligocene.

This rotation is significantly larger than the 38° reported by Horner and Freeman [12] from a study of Palaeocene and Eocene sublithographic limestones in the same zone. Such a straightforward comparison, which would imply an 11^0 anticlockwise rotation between Upper Eocene and Lower Oligocene, is however delicate: First of all, Horner and Freeman have interpreted their data with respect to the Early Tertiary pole position of Van der Voo and French [27] which differs significantly from the one given by Tauxe et al. [26]. The inclinations obtained by Horner and Freeman agree precisely with the Eocene pole of Van der Voo and French, but are steeper than expected on the basis of the one given by Tauxe et al. [26]. Because limestones are known to yield faithful paleomagnetic directions and inclinations steeper than expected are difficult to explain, it seems reasonable to us to use Van der Voo and French's result for the Eocene as the authors have done. The results of Van der Voo and French, however, do not extend into the Oligocene, so that our results are referred to the mentioned pole of Schult [25], which is the only one existing for the Oligocene. This is clearly an unsatisfactory compromise, due to the lack of reliable and coherent Tertiary polar wander paths for Africa.

Moreover, the results of Horner and Freeman only acquire a regional significance after an orocline correction obtained by rotating the sampled structures in order to align them with the present regional trend, estimated at N150° [12]. In our opinion, however, a better estimation of the regional trend is N155°, as given by the strike of the axis of the Botzara Syncline, which is the major structure of Epirus. This estimation would contribute an additional 5° to the rotation, which is furthermore increased by 1° when a slightly erroneous appreciation of the strike of the axis of the Dodoni Syncline is taken into account. On this premise the results of Horner and Freeman may actually indicate a rotation of 44° and not 38°. Alternatively if the Botzara Syncline is not considered as indicating the regional trend, an orocline correction must be performed on our results from the flysch sections which then indicate a clockwise rotation of $44.5 \pm 6.4°$.

A statistical test [28] shows that, taking into account this correction and the possible inclination error, there is no significant difference between the two results: if an anticlockwise rotation has occurred between Upper Eocene and Lower Oligocene, its maximum amplitude is of the order of 5°. Although the above discussion shows that

comparison of the two polar paths is not easy, it is worthwhile to note that an anticlockwise rotation of this amplitude is obtained from the path of Tauxe et al. [26].

In summary these two studies, obtained from different lithologies, agree on a clockwise rotation of about 45°, which is considerably larger than the 26° reported from Upper Miocene and Plio-Quaternary sediments in the Ionian Islands [10].

This recent rotation started at about the Mio-Pliocene boundary, after a period extending to at least the Serravallien-Tortonian boundary during which no significant rotation has occurred [5,10]. Because there are no major geological or tectonic discontinuities between the Ionian Islands and mainland Greece, this rotation has certainly affected the Ionian Zone, but only accounts for about one half of the rotation observed in the flysch sections. Thus the paleomagnetic results suggest that northwestern Greece has undergone two distinct clockwise rotations, of about 20–25° each, one between Lower to Middle Miocene and one during the Pliocene and Quaternary, while no other major Tertiary rotation seems to have occurred earlier. The lack of suitable outcrops in the 24–12 My period mentioned above does not allow more accurate determination of the period of activity of the Miocene rotation. This situation is shown schematically in Fig. 10.

These two phases of rotation can be interpreted within the framework of known geological observations. Mercier et al. [4] have shown that an important compressional phase has affected the Ionian Islands in the Lower Pliocene and suggested that this tectonic phase marks the initiation of the external Aegean Arc as an active continen-

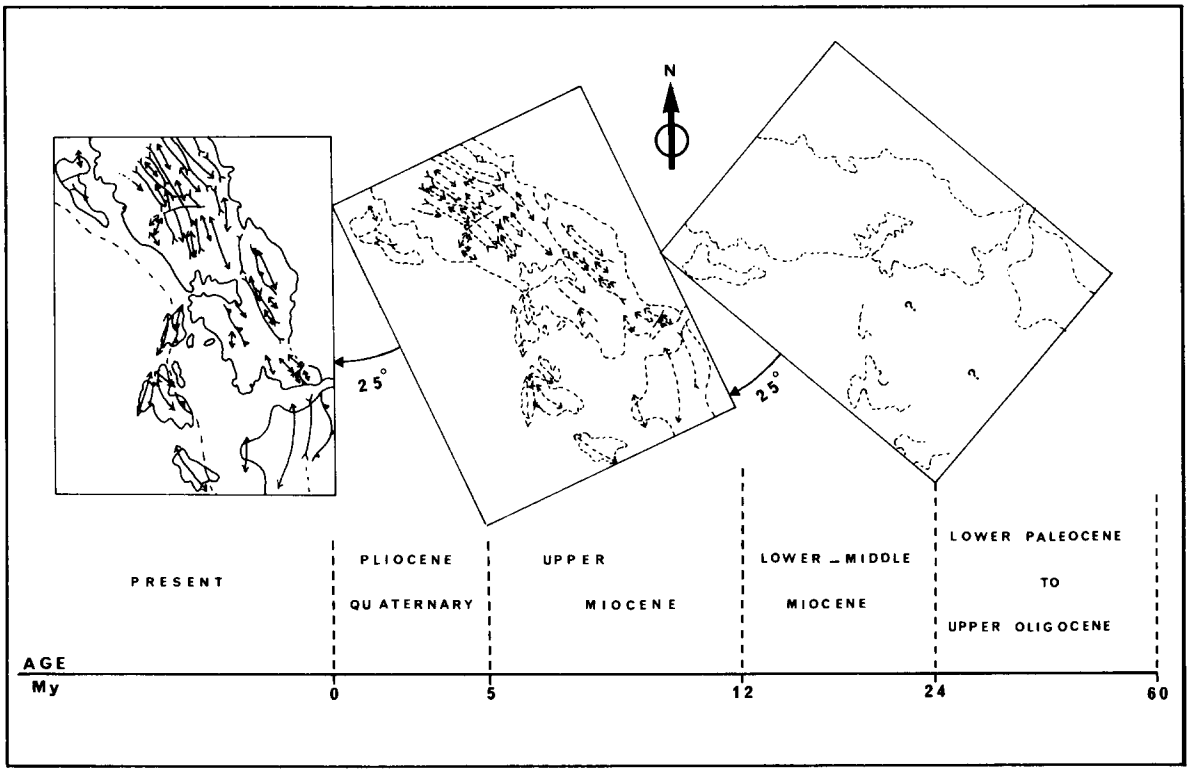


Fig. 10. Clockwise rotations of northwestern Greece since Lower Miocene. Ionian structures are not represented in the map corresponding to Lower Miocene because they were not formed at this epoch. In the same way, the coastal outlines are drawn like they are today at Lower Miocene and Plio-Quaternary limit only to supply a visual support to the rotation.

tal margin under compression. It has already been remarked in a previous paper [10] that the beginning of the Plio-Quaternary rotation overlaps in time with the Lower Pliocene compressive phase, thus yielding support to this interpretation. Furthermore we note that the already mentioned last phase of the "Alpine" tectonic activity in this region occurred at the Late Burdigalian/Early Langhian stage. We thus suggest that the Miocene rotation is related to this major tectonic activity, so that the two rotations are directly linked to the known structural history of this region.

An important geological consequence of the measured rotation is that it implies a large horizontal displacement whose minimum amount can be simply estimated. Indeed, the compressional tectonics and general westward directed thrusting necessarily imply that the Eulerian pole is situated to the north of the Ionian Zone. By placing a pivot point immediately north of the studied region, the minimum displacement corresponding to a 45° rigid rotation can then be evaluated as 130 km in southern Epirus and 240 km in southern Akarnania. Because the investigated structures extend much farther in Albania the real displacement is certainly greater.

The question then arises as to whether the sedimentary cover alone or the basement as well is affected by this displacement.

Extensive low-angle thrusting involving the sedimentary cover has been recognized in western Greece and in Peloponnesos by several authors for both the Lower to Upper Miocene and Plio-Quaternary periods [1,13,29–32]. Moreover, Aubouin has recognized a general increase of the allochtony of the structures from the lateral toward the central parts of the arc [32]. This implies a general southward progressive increase of the horizontal displacements, in qualitative agreement with the existence of a clockwise rotation about a Eulerian pole situated in the northwestern region. Furthermore, the amplitude of the displacement evidenced by the presence of windows in Peloponnesos [33] amounts to a few hundreds kilometers, which is of the same order of magnitude as the one obtained in southern Akarnania from the paleomagnetic results. Therefore this general pattern is compatible with a paleomagnetic rotation associated with the movement of the nappe sheets.

Other evidence, however, seems to indicate that the basement itself is involved in the rotation. First of all, because there is no evidence of gravitational sliding, it has been recognized, as early as 1960, that the cover deformation could arise from lithospheric shortening [34].

The paleomagnetic results form Epirus and Akarnania are very coherent in spite of the already mentioned large differences in the tectonic cover of these two regions. In addition, results obtained from sites located in the Ionian and Preapulian zones in the Ionian Islands, and deposited before the thrust occurred, are identical [5,10]: at least in this case, the movement of the nappe sheet has not introduced any differential paleomagnetic rotation.

The paleomagnetic results could thus be equally interpreted assuming a rotation not limited to the sedimentary cover, over a decollement level distinct and deeper than the one over which the thrust of the Ionian Zone over the Preapulian Zone took place. Because the Plio-Quaternary rotation has occurred in a zone of continental collision, the hypothesis of large scale shortening seems more realistic in this case, in our opinion. However, the present results are not exhaustive and precise enough to yield a clear cut interpretation for the Miocene rotation. The total measured rotation could arise from a combination of mechanisms involving both rotation of the nappe sheets and rotation of the basement as already suggested [12].

Another geological phenomenon, the large calc-alkaline volcanic activity in the Aegean Sea, seems to be related to the tectonic activity leading to the two phases of rotation measured in northwestern Greece. Several authors [35–37] have indeed shown that the calc-alkaline activity consists of two main phases, one during the Oligo-Miocene and one during the Plio-Quaternary stages, separated by a quiescent period of about 8 My. The Oligocene products are found in Thrace (34–24 My). Then starting in Lower Miocene a southward migration of the volcanic activity has been recognized with the ages becoming progressively younger up to 13.2 My in central Evia. This southward migration is thus synchronous with the

Miocene rotation in northwestern Greece. The calc-alkaline activity then came to an end. It only restarted about 5 My ago and is still active today. Thus there appears to be a time correlation between the two phases of rotation and the two phases of calc-alkaline volcanic activity. Moreover the direction of the southward migration is the one expected assuming, as is the case, a clockwise rotation about a pole situated to the west of the area affected by the volcanism. Such a relation between volcanic and tectonic activity has already been suggested by Bellon et al. [36].

The most important geological consequence of these paleomagnetic results concerns the acquisition of the curvature of the arc. Using the paleomagnetic data to rotate the region back to its original orientation, one finds that the Lower Miocene regional trend was almost east-west (N110°) when the Ionian structures have been formed. At the other termination of the arc, in southern Anatolia, the Langhian tectonic activity was also very large and ended with the emplacement of the Lycian nappes. The presence of windows in these nappes has permitted Poisson [38] to obtain evidence of a progressive increase of the southward displacement of the nappes when moving westward from Isparta to the coast. An apparent anticlockwise rotation of the nappe front has been postulated by Poisson [38] who also concluded to an east-west trend of the Lycian structures at Lower Miocene. It must be noted however that the paleomagnetic results obtained by Lauer [39] and Kondopoulou and Lauer [40] for the Miocene period in central Aegea and western Turkey indicated a clockwise rotation of all these regions in disagreement with Poisson's results. Unfortunately, these data are too scarce, particularly in western Turkey (only 3 sites) and too scattered to give a definitive conclusion. Assuming that the geological results of Poisson are better documented, we notice that there is a coherence between the Lower Miocene orientations at the two terminations of the arc. It indicates that the structure of the Lower Miocene arc was nearly rectilinear. So that the curvature observed today has been almost entirely acquired due to its tectonic activity.

As the curvature has been generally considered as already existing in Eocene times, this result appears to constitute a new element for a better understanding of the geodynamical evolution of the Eastern Mediterranean.

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